

1. Configure Classic Load balancer.

- ➔ Created and configured a Classic Load Balancer (CLB) in AWS.
- ➔ Configured the Load Balancer as internet-facing.
- ➔ Selected the required VPC and availability zones.
- ➔ Configured the listener to accept HTTP traffic on port 80.
- ➔ Configured the required security group rules for inbound traffic.
- ➔ Registered the required EC2 instances with the Load Balancer.
- ➔ Configured health checks to monitor the availability of the registered instances.
- ➔ Verified that the registered instances were healthy.
- ➔ Tested the application through the Classic Load Balancer DNS name.

Your VPCs

Filter by VPC: ▼

AWS Global View [↗](#)

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways

Your VPCs (2) Info

Find VPCs by attribute or tag

Last updated 2 minutes ago [Actions](#) [Create VPC](#)

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block P
☐	-	vpc-0f3216a7937311169	Available	-	-	Off
☐	LB-Lab-VPC	vpc-02635a0bc2ca08952	Available	-	-	Off

Internet gateways (1+) Info

Find internet gateways by attribute or tag

[Actions](#) [Create internet gateway](#)

☐	Name	Internet gateway ID	State	VPC ID	Over
☐	Name tag: LB-Lab-IGW	igw-039846a6eb85cdd0d	Attached	vpc-02635a0bc2ca08952 LB-Lab-VPC	95

Subnets (2) Info

Find subnets by attribute or tag

Last updated 1 minute ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...
☐	LB-Public-Subnet-1	subnet-0d96645d3130d2337	Available	vpc-02635a0bc2ca08952 LB-L...	Off
☐	LB-Public-Subnet-2	subnet-0fa6613d6083251f4	Available	vpc-02635a0bc2ca08952 LB-L...	Off

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EC2 > Instances

EC2

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 - Volumes
 - Snapshots

Instances (1/1) Info

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4
LB-Web-Serve...	i-04d754d5a9f8245da	Running	t3.micro	Initializing	us-east-1a	-

i-04d754d5a9f8245da (LB-Web-Server-1)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-04d754d5a9f8245da

IPv4 address 44.193.210.223 | open address

Instance state Running

Private IPv4 addresses 10.0.1.244

Public DNS -

Public IPv4 address copied

← → ↺ ⚠ Not secure 44.193.210.223

Classic Load Balancer - Server 1

AZ: us-east-1a

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Instances (1/2) Info

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public
LB-Web-Server-2	i-00394fd60aabb8335	Running	t3.micro	Initializing	us-east-1b	-
LB-Web-Server-1	i-04d754d5a9f8245da	Running	t3.micro	3/3 checks passed	us-east-1a	-

i-00394fd60aabb8335 (LB-Web-Server-2)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID i-00394fd60aabb8335

Public IPv4 address 3.82.125.110 | open address

Instance state Running

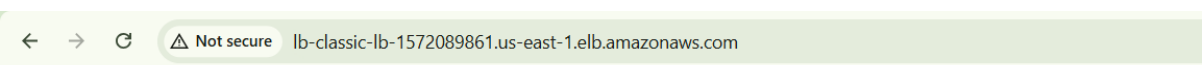
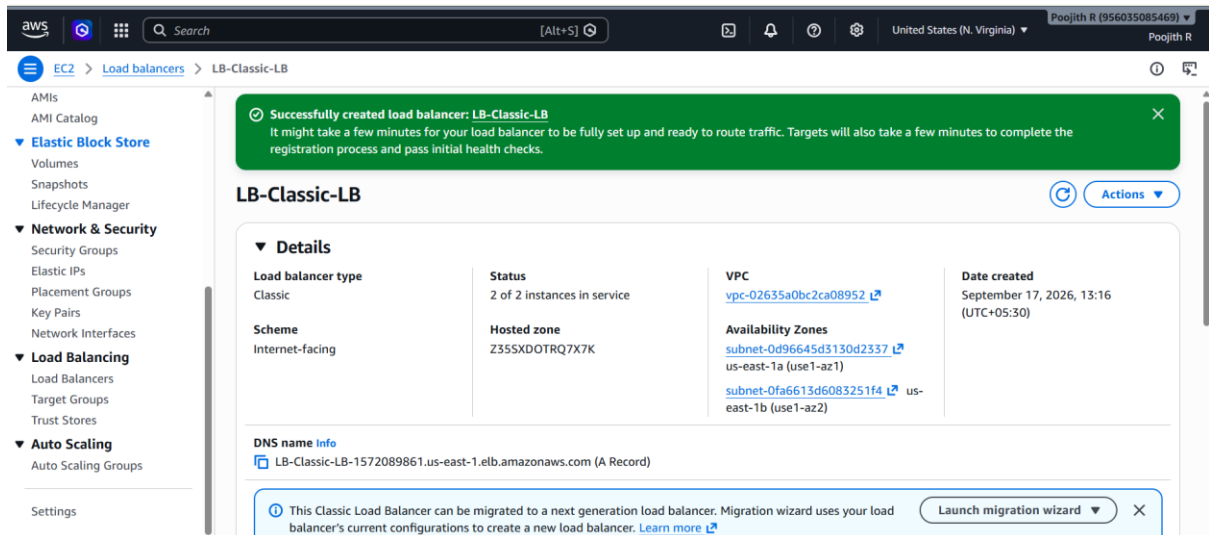
Private IPv4 addresses 10.0.2.132

Public DNS -

← → ↺ ⚠ Not secure 3.82.125.110

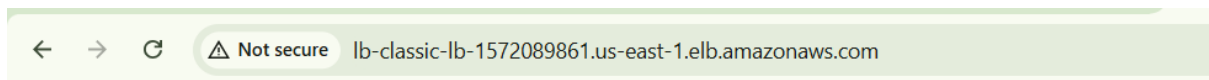
Classic Load Balancer - Server 2

AZ: us-east-1b



Classic Load Balancer - Server 2

AZ: us-east-1b



Classic Load Balancer - Server 1

AZ: us-east-1a

2. Configure Application Load balancer.

- ➔ Created an Application Load Balancer (ALB) named LB-Application-LB.
- ➔ Configured the ALB as internet-facing with IPv4 connectivity.
- ➔ Selected multiple Availability Zones for high availability.
- ➔ Created/configured the target group LB-ALB-TG.
- ➔ Registered the required EC2 instances as targets.
- ➔ Configured the HTTP listener on port 80.
- ➔ Configured the listener to forward requests to the target group.
- ➔ Verified that the registered EC2 targets were healthy.
- ➔ Tested the application using the ALB DNS name.

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EC2

Target groups

LB-ALB-TG

EC2

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Lifecycle Manager

LB-ALB-TG

Actions

Details

arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/LB-ALB-TG/dc8783297cd03bd9

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-02635a0bc2ca08952

IP address type

IPv4

Load balancer

None associated

2

0

0

2

0

0

Total targets

Healthy

Unhealthy

Unused

Initial

Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

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Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

LB-Application-LB

Actions

Successfully created load balancer: LB-Application-LB

It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

Introducing new Capacity Utilization metric for ALB

Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the Capacity or Monitoring tabs to view utilization data.

Details

Load balancer type

Application

Status

Provisioning

VPC

vpc-02635a0bc2ca08952

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35SXD0TRQ7X7K

Availability Zones

subnet-0d96645d5130d2337 us-east-1a (use1-az1)

subnet-0fa6613d6083251f4 us-east-1b (use1-az2)

Date created

September 17, 2026, 13:28 (UTC+05:30)

Load balancer ARN

DNS name Info

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LB-ALB-TG

Actions

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

1

	Instance ID	Name	Port	Zone	Health status	Health status details	Admin
☐	i-00394fd60aabb8335	LB-Web-Server-2	80	us-east-1b (us-east-1a)	Healthy	-	None
☐	i-04d754d5a9f8245da	LB-Web-Server-1	80	us-east-1a (us-east-1b)	Healthy	-	None

CloudShell

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Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the **Capacity** or **Monitoring** tabs to view utilization data.

LB-Application-LB

Details

Load balancer type Application	Status Active	VPC vpc-02635a0bc2ca08952	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0d96645d3130d2337 us-east-1a (use1-az1) subnet-0fa6613d6083251f4 us-east-1b (use1-az2)	Date created September 17, 2026, 13:28 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/LB-Application-LB/b18b5265960fcc3		DNS name Info LB-Application-LB-1220850204.us-east-1.elb.amazonaws.com (A Record)	

← → ↻ ⚠ Not secure lb-application-lb-1220850204.us-east-1.elb.amazonaws.com

Classic Load Balancer - Server 1

AZ: us-east-1a

← → ↻ ⚠ Not secure lb-application-lb-1220850204.us-east-1.elb.amazonaws.com

Classic Load Balancer - Server 2

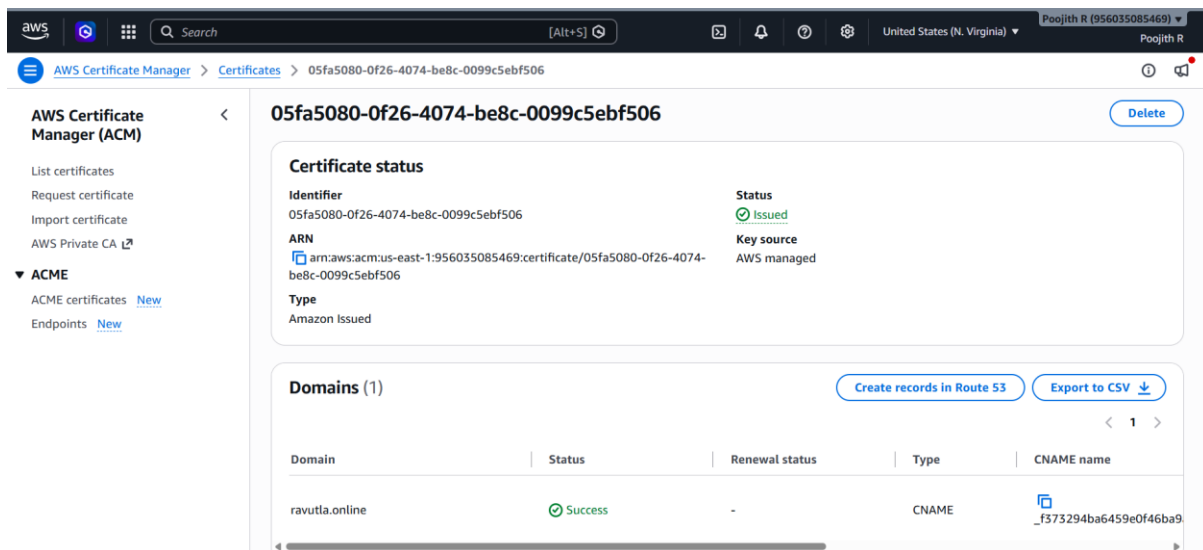
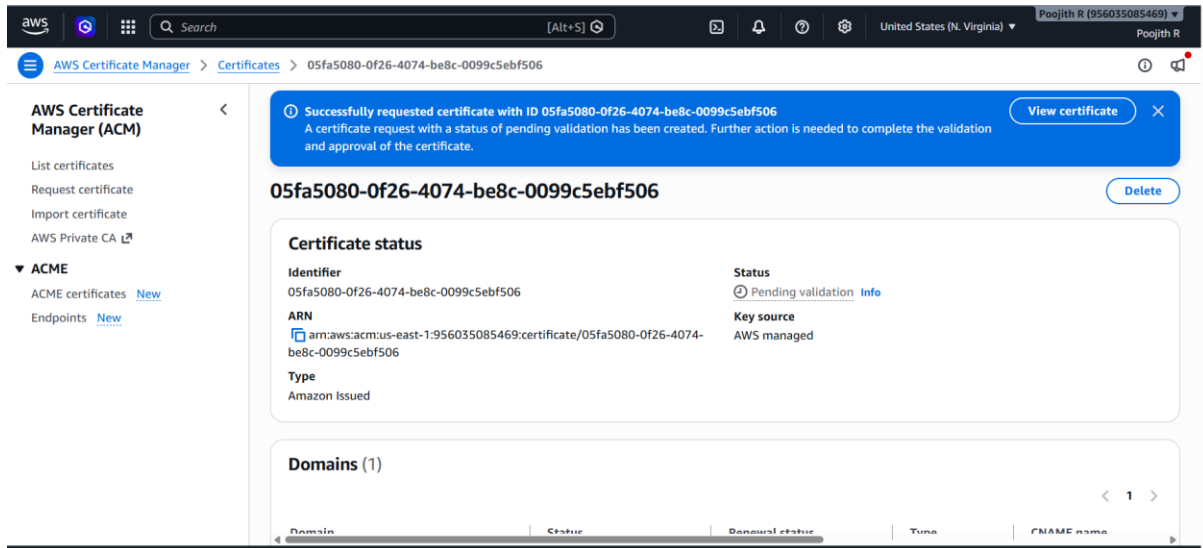
AZ: us-east-1b

3. Configure Network Load balancer.

4. Attach SSL for application load balancer.

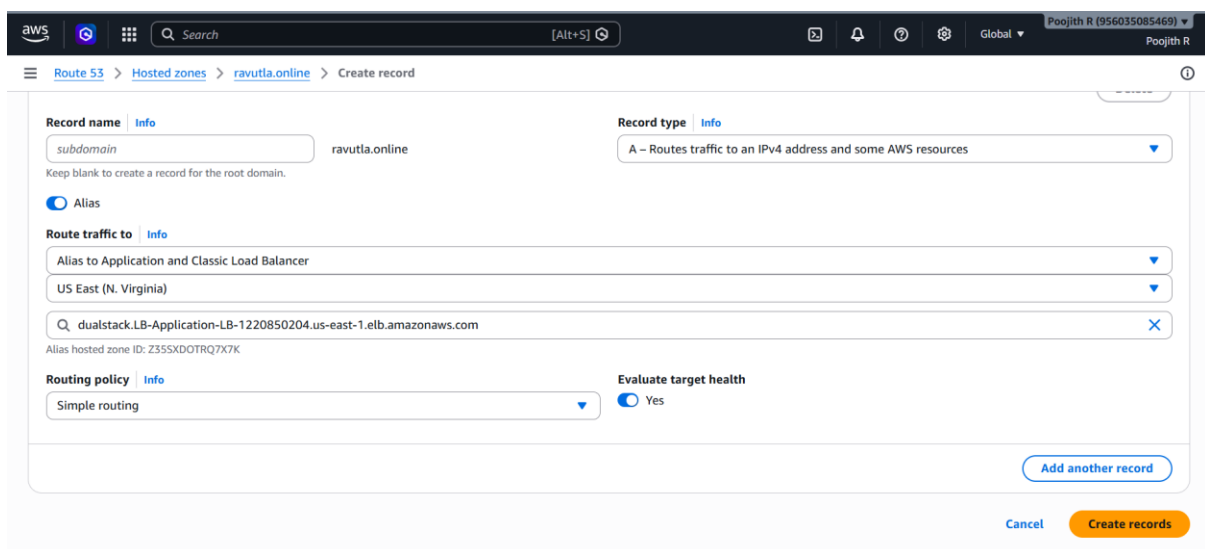
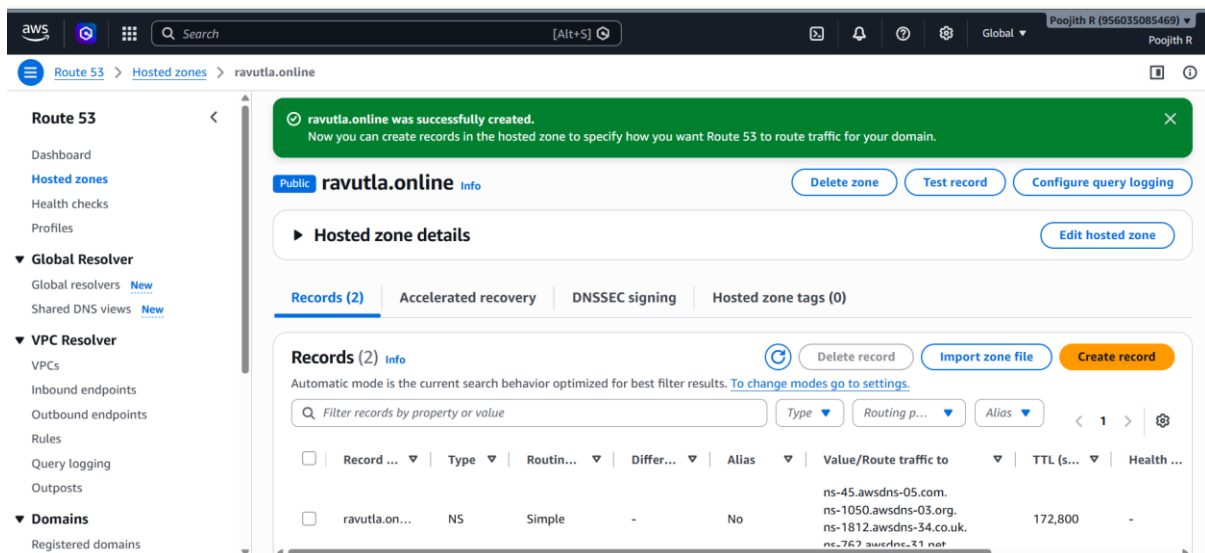
- ➔ Created/requested an AWS Certificate Manager (ACM) certificate for ravutla.online.
- ➔ Completed DNS validation for the domain.
- ➔ Verified that the ACM certificate status was Issued.
- ➔ Configured an HTTPS listener on port 443 on the Application Load Balancer.

- ➔ Attached the ravutla.online ACM certificate to the HTTPS listener.
- ➔ Configured HTTPS traffic to forward to LB-ALB-TG.
- ➔ Added an inbound HTTPS/TCP 443 rule to the ALB Security Group.
- ➔ Tested secure access using:
https://ravutla.online
- ➔ Verified that the website was accessible over HTTPS with a valid certificate.



5. Map Application load balancer to R53.

- ➔ Created/configured the public hosted zone ravutla.online in Amazon Route 53.
- ➔ Updated the domain's nameservers at GoDaddy to use the Route 53 nameservers.
- ➔ Created an A record with Alias in Route 53.
- ➔ Configured the Alias record to point to the Application Load Balancer.
- ➔ Verified the ALB DNS target as:
dualstack.lb-application-lb-1220850204.us-east-1.elb.amazonaws.com
- ➔ Verified DNS propagation using nslookup.
- ➔ Confirmed that ravutla.online resolved to the ALB.
- ➔ Tested both HTTP and HTTPS access through the custom domain.
- ➔ Successfully accessed the application using https://ravutla.online.



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Windows\System32>
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Windows\System32> nslookup -type=NS ravutla.online
Server: 192.168.0.1
Address: 192.168.0.1

Non-authoritative answer:
ravutla.online nameserver = ns-1050.awsdns-03.org
ravutla.online nameserver = ns-45.awsdns-05.com
ravutla.online nameserver = ns-762.awsdns-31.net
ravutla.online nameserver = ns-1812.awsdns-34.co.uk

ns-45.awsdns-05.com internet address = 205.251.192.45
ns-45.awsdns-05.com AAAA IPv6 address = 2600:9000:5300:2d00::1
PS C:\Windows\System32> nslookup ravutla.online
Server: 192.168.0.1
Address: 192.168.0.1

Non-authoritative answer:
Name: ravutla.online
Addresses: 3.216.35.116
          54.198.67.207

PS C:\Windows\System32> |
```

```
< > ↻ 🌐 ravutla.online
```

Classic Load Balancer - Server 2

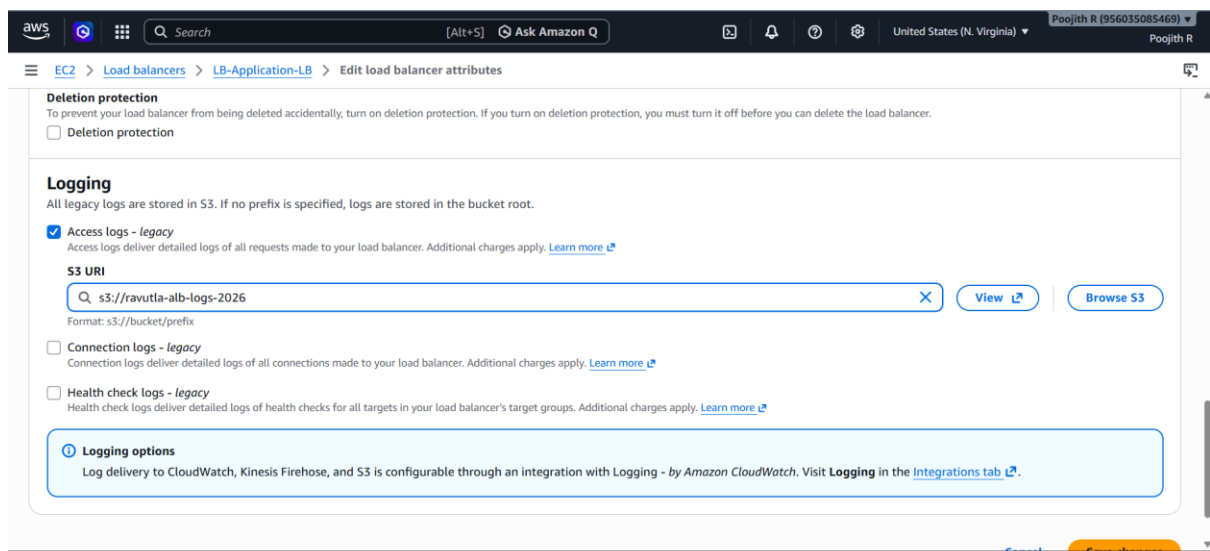
AZ: us-east-1b

Classic Load Balancer - Server 1

AZ: us-east-1a

6. Push the application load balancer logs to S3.

- ➔ Created a dedicated S3 bucket named ravutla-alb-logs-2026 in us-east-1.
- ➔ Kept Block Public Access enabled for the bucket.
- ➔ Configured the required S3 bucket policy to allow AWS Load Balancer log delivery.
- ➔ Enabled ALB Access Logs.
- ➔ Configured the S3 destination as:
s3://ravutla-alb-logs-2026
- ➔ Verified the AWS ELBAccessLogTestFile was successfully created.
- ➔ Generated traffic to https://ravutla.online to produce ALB access logs.
- ➔ Verified the actual ALB log files were delivered to S3.
- ➔ Downloaded and opened the generated ALB log file to verify successful log delivery.



Successfully modified load balancer attributes.

Introducing new Capacity Utilization metric for ALB
Monitor how much of your load balancer's available capacity was used to meet traffic demand. Visit the **Capacity** or **Monitoring** tabs to view utilization data.

LB-Application-LB

Details

Load balancer type Application	Status Active	VPC vpc-02635a0bc2ca08952	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-0d96645d3130d2337 us-east-1a (use1-az1) subnet-0fa6613d6083251f4 us-east-1b (use1-az2)	Date created September 17, 2026, 13:28 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:956035085469:loadbalancer/app/LB-Application-LB/b18b5265960fcc3		DNS name info LB-Application-LB-1220850204.us-east-1.elb.amazonaws.com (A Record)	

Amazon S3 > ... > ravutla-alb-logs-2026 > AWSLogs/ > 956035085469/ > elasticloadbalancing/ > us-east-1/ > 2026/ > 09/ > 17/

17/

Copy S3 URI

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
956035085469_elasticloadbalancing_us-east-1_app-LB-Application-LB/b18b5265960fcc3_20260917T1020Z_3.216.35.116_2v71mtx5.log.gz	gz	September 17, 2026, 15:50:10 (UTC+05:30)	1.2 KB	Standard

956035085469_elasticloadbalancing_us-east-1_app-LB-Application-LB/b18b5265960fcc3_20260917T1020Z_3.216.35.116_2v71mtx5.log.gz (evaluation copy)

File Commands Tools Favorites Options Help

Extract To Test View Delete Find Wizard

956035085469_elasticloadbalancing_us-east-1_app-LB-Application-LB/b18b5265960fcc3_20260917T1020Z_3.216.35.116_2v71mtx5.log.gz

Name	Size	Packed	Type
956035085469_e...	9,265	1,195	Text Document

Selected 1 file, 9,265 bytes

Curvy Fonts.txt protest - nirasana.txt Benzo Vs EM.txt Untitled IAM_Notes 95603508546

```
h2 2026-09-17T10:18:24.879598Z app/LB-Application-LB/b18b5265960fcc3 103.203.172.230:49273 10.0.2.132:80 0.001 0.001 0.000 200 200 653 267 "GET https://ravutla.online:443/ HTTP/2.0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36" TLS_AES_128_GCM_SHA256 TLSv1.3
arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/LB-ALB-TG/dc8783297cd03bd9 "Root=1-6aabb7e0-78b89c041958867a727341dc" "ravutla.online" "arn:aws:acm:us-east-1:956035085469:certificate/ab3048b7-1807-44cd-b8d5-1bca0d32eef8" 0 2026-09-17T10:18:24.878000Z "forward" "-" "-" "10.0.2.132:80" "200" "-" "-"
TID_ad4d909973fc7741bf529b440549dcb4 "-" "-" "-" 3.216.35.116 "-" "-"
h2 2026-09-17T10:18:29.276235Z app/LB-Application-LB/b18b5265960fcc3 103.203.172.230:49273 10.0.2.144:80 0.000 0.000 0.000 200 200 66 265 "GET https://ravutla.online:443/ HTTP/2.0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36" TLS_AES_128_GCM_SHA256 TLSv1.3
arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/LB-ALB-TG/dc8783297cd03bd9 "Root=1-6aabb7e5-54443a2d64b362b84fe43776" "ravutla.online" "arn:aws:acm:us-east-1:956035085469:certificate/ab3048b7-1807-44cd-b8d5-1bca0d32eef8" 0 2026-09-17T10:18:29.275000Z "forward" "-" "-" "10.0.1.244:80" "200" "-" "-"
TID_ad4d909973fc7741bf529b440549dcb4 "-" "-" "-" 3.216.35.116 "-" "-"
h2 2026-09-17T10:18:30.226765Z app/LB-Application-LB/b18b5265960fcc3 103.203.172.230:49273 10.0.2.132:80 0.001 0.001 0.000 200 200 27 266 "GET https://ravutla.online:443/ HTTP/2.0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36" TLS_AES_128_GCM_SHA256 TLSv1.3
arn:aws:elasticloadbalancing:us-east-1:956035085469:targetgroup/LB-ALB-TG/dc8783297cd03bd9 "Root=1-6aabb7e6-6f17bd343bc77ee91fedd57f" "ravutla.online" "arn:aws:acm:us-east-1:956035085469:certificate/ab3048b7-1807-44cd-b8d5-1bca0d32eef8" 0 2026-09-17T10:18:30.225000Z "forward" "-" "-" "10.0.2.132:80" "200" "-" "-"
TID_ad4d909973fc7741bf529b440549dcb4 "-" "-" "-" 3.216.35.116 "-" "-"
h2 2026-09-17T10:18:30.991158Z app/LB-Application-LB/b18b5265960fcc3 103.203.172.230:49273 10.0.2.144:80 0.000 0.000 0.000 200 200 27 265 "GET https://ravutla.online:443/ HTTP/2.0" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)"
```